An Empirical study on Stability of Beta in Indian Stock Market with special reference to CNX NIFTY FIFTY

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ABSTRACT

This paper investigates the stability of beta in Indian stock markets using fifteen years of daily data of CNX NIFTY FIFTY from 2000 to 2015. The CAPM beta was computed by using market model regression. Chow breakpoint test was used to examine the impact of the 2008 subprime crisis on the stability of beta. Unknown breakpoints were investigated in the beta series using multiple breakpoint tests on both individual stocks and portfolios. Subsequently CUSUM was adopted to check for the sequential changes in the computed beta series. The results indicate that 2008 subprime crisis has not much influenced on structure of the beta series. However, we have found a few unknown break points in different time periods. Our results show that the betas vary across the study time periods and portfolio betas as well as those of the individual stocks. The beta stability plays an important role while estimating the portfolio returns, the individual stock returns and in devising winning strategies. Therefore, it is highly recommended to the market participants while using historical betas for prediction of future risk of the stock and portfolio need to be extra cautious.

Key Words: Multiple breaks, CUSUM test, CNX Nifty 50, Subprime Crisis, Unknown breaks

I. INTRODUCTION

The two key determinants for holding any financial instrument is its expected reward (return) and risk. Rational investors' prefer those financial instruments that have low risk and high returns. That means the securities that have more market risk should offer higher returns to entice the investors. Market risk derives from the accidental changes in the prices of financial assets. How to predict market risk is very vital for the market participants. In common parlance, risk is the chance of financial loss. Assets having greater chances of loss are viewed as more risky than those with lesser chances of loss. Therefore one can define risk as *the variability of the actual return from the expected returns associated with a given asset/investment.* The greater the variability, the riskier the security (e.g. shares) is said to be.

The modern theory of portfolio analysis dates back to the seminal work of Harry Markowitz. In this theory Markowitz advocated that all rational investors select among portfolios on the basis of two parameters, that is the expected risk (variance) and the expected return associated with that portfolio. Subsequently, the capital asset pricing model of Sharpe (1964), Jan Mossin (1966) and Lintner (1965) were responsible for the evolution of asset pricing model. CAPM is a model for establishing optimal portfolios. In this model they proposed that a stock's risk is characterized by its systematic risk BETA. Beta measures a stock's volatility, the degree to which stock's price varies in relation to the overall market movement. Securities' betas have been estimated for various individual stocks and for evaluating the performance of well diversified portfolios by the market participants. The beta of a firm can be estimated by regressing (OLS) its rate of return against the market portfolio return. CAPM is still a widely accepted and used model for the estimation of the cost of capital of a firm.

Apart from this, the theory states that, beta has a predictive power of prospective risk of a stock and it helps the investigator to estimate exactly the security's prospective riskiness. Therefore as per CAPM, beta coefficient is a key parameter, which helps in determining the minimum expected rate of return on risky stocks, construction of portfolio and performance evaluation. As future beta is calculated from historical data, the stability of beta with time horizon is very crucial for the market participants so that the prediction is true and appropriate. However, the anomalies that exist in the financial market, trade cycles, macro-economic factors etc. influence the beta value. Therefore Beta stability and sturdiness evaluations have become one of the prime focuses of research in finance.

The systematic risk of a stock or a portfolio can be determined by its historical returns. However, CAPM theory fails to state any ground rules on which return interval and estimation period should be chosen for the computation of beta, that is whether rates of return should be measured over a day, a week or a year. Generally, daily returns are used for the estimation of beta. If returns are independently distributed, betas computed using high frequency returns (daily) should not be considerably different from a low frequency data (ex. Weekly, or monthly returns). But empirical findings of Dimson & Marsh (1983) , Hawawini (1983), Cohen et al., (1986) and Draper and Paudyal (1995), Kim, (1999) show that beta estimates on the basis of daily, weekly, monthly or yearly rates of return are significantly differing from each other, although they are calculated for the same period of time. They concluded that betas based on daily returns bring about more stable beta than those calculated using low frequency data.

One of the major observations documented in literature is by Kon & Jen, (1978; 1979) they noted that the stock market reacts differently during bull and bear phases. This would result in different betas for different phases even if the beta coefficient had a stable bull and bear value. Similar findings are documented by the subsequent researchers like Fabozzi and Francis (1977), Wiggins (1992), Chen (1982) Bhardwaj and Brooks (1993).

One such possible reason is that time-scale dependence of systematic risk of stocks or assets. In his study Dubey (2014) documented that stocks exhibit considerable instability in their beta estimates as far as different investment horizons are concerned because of tremendous heterogeneity with different time horizon in investments. Apart from this he argued that in emerging market economies, conditions are very fluid.

Beta is related to the leverage of the firm. The systematic risk, which is the risk inherent to the entire market or entire market segment of a share, can be further divided into two components. First component is operational risk and the other is financial risk. According to Hamada, (1972); Mandelker & Rhee, (1984) financial risk is a function of the leverage of the firm. As leverage can change over a period of time which leads to changes in stock price movements, that leads to changes in beta of the security or the portfolio. Lally (1998) pointed out that controlling for the degree of financial leverage may improve beta forecasting. Jagannathan and Wang (1996) and Lewellen and Nagel (2003), McNulty et al., (2002), (Choudhry and Wu, 2009) documented that stock market conditions (the volatility and causes of volatility) affect the leverage of the firm and hence the beta of the firm and that is the major reason behind the instability of the beta over the period of time. Any new information that does not affect both stock index and individual stock returns homogeneously, would change the relationship (correlation) between the two and automatically beta of the stock would change.

Sometime instability of beta may occur because of measurement error. According to Alexander & Chervany (1980) theoretical beta relates before the event expectations while estimated beta relates after the event observations. Scott and Brown (1980) documented that auto correlation in the residuals is also responsible for the instability in Beta estimation.

The beta stability has been extensively studied in the western financial literature. The literature has applied a range of models on individual stocks as well as portfolios to check the stability of beta. Several studies tries to investigate the nature and the behavior of beta in terms of intervals taken for computation on beta stability, modus operandi for computation of beta on stability of beta. Couple of studies tries to investigate stability of portfolio beta with individual stock beta, stability of beta over time and with different market conditions such as bull and bear phases. However, there is no clear consensus about the beta stability on individual stocks and portfolios.

As stated earlier, the estimation of beta is vital for many applications in finance. The research on Indian markets in regard to beta stability has been relatively few as compared with western markets. However, in reviewing the Indian literature, the evidence on stability beta of Indian capital markets have also showed mixed response. Moreover, the beta estimation is more sensitive issue in developing economy like India as Indian equity markets witnessed sensational changes in risk and returns profile of stocks because of fast changing trends in regulation and reforms, volatile business cycle, etc. However, research on beta stability in emerging markets is limited in the literature. Thus the main objective of this paper is to investigate the stability of betas in Indian stock market with special reference to CNX Nifty Fifty stocks for a period of fifteen years during 2000 to 2015. Thus, the current study, contributes to the existing literature by taking into 2008 subprime crisis and its impact on the beta stability and testing the stability of beta by using models like multiple breakpoints and CUSUM tests in Indian security market.

Objectives of the study

The current study tries to answer the following objectives:

1. To identify whether there is any structural break in the computed beta series with reference to subprime crisis 2008;
2. To investigate any unknown breaks in the computed beta series, and
3. To examine the behavior and stability of beta across stocks listed in CNX Nifty Fifty

II. LITERATURE REVIEW

A number of empirical studies have tried to investigate the stability of beta across the stock markets. Blume (1971) was the first researcher who investigated on the beta stability who shows empirically that the stability of stock beta increases with the increase in the time of estimation period. Subsequent studies have addressed the question of beta's stability over time. The most prominent being Baesel (1971) who concluded that "the stability of beta is reliant on two factors; first factor is the estimation period used and second factor is boundary of the beta picked". Moreover, he claims that the optimal estimation period is nine years. However Gonedes claims the optimal estimation period is seven years. Gordon and Chervany (1980) contradicted the Baesel's claim of longer the estimation duration the more stable the betas. They concluded that the ideal estimation interval could be four to six years.

Levy (1971) has documented that the portfolio betas are stable while individual security betas are unstable., Altman, Jacquillat, and Levasseur (1974), Roenfeldt et al., (1978), Alexander and Chervany (1980) and Theobald (1981), Sunder (1980), Bos and Newbold, (1984); Faff et al.,(1992), Kok (1994); Cheng (1997); Chawla (2001), Moonis and Shah, (2002) and Cheng and Boasson (2004) supported this view that beta is far from being stable. Alexander and Benson (1982) found that stocks had varying betas. Bos and Newbold (1984), Collins et al. (1987) found that 58% of stocks had varying betas and conclude that the beta is not constant over time. Vasicek (1973) found that the longer the estimation periods, the more stable the estimates become. Vipul (1999) concluded that the stability of beta varies based on size and liquidity of the firm.

Many empirical studies have been conducted to investigate the relationship between conditions of the stock market and systematic risk by measuring beta including Black (1972), Francis (1975) Brenner and Smidt (1977) Fabozzi and Francis (1977), Kim and Zumwalt (1979), DeBondt and Thaler (1987), Spiceland and Trapnell (1983) and Wiggins (1992)who documented that beta varies with market conditions. In a study by Xia Pan, et al., (2014) to investigate the stability of beta in Chinese Stock Market concluded that Macroeconomic factors such as general business environment and big world events greatly affect company's beta. However, microeconomic factors like merger and acquisitions, change in management etc. hardly have any impact on betas. In a study by Hakan Er and Sevgi Aydin estimated the betas on the basis of daily, weekly, bi-weekly and monthly returns of 225 companies listed in ISE over a period of nine years revealed that interval and length of the estimation period have a major impact on the estimates of beta.

In a study by Levy (1971), Blume (1971), Levitz (1974) and Altman (1974) to examine the stability of individual stocks as well as portfolio betas concluded that portfolio betas are more stable than individual stock betas.

In an investigation by Thomas and George (2010) to examine the stationarity of beta for a period of 14 years and three sub periods 27 out of 60 companies beta were unstable. They concluded that beta varies in accordance with the vagaries of time. In an investigation by Elton Scott and Stewart Brown (1980) on the stability of betas concluded that autocorrelated residuals and intertemporal market-residual correlations between residuals and market returns can result to unstable and biased estimates of betas. In an investigation by Mahmoud Haddad et al., (2008) to examine the Beta stationarity of social Islamic mutual funds performance during 1997 to 2002 concluded that various sector Islamic mutual funds displays time varying volatility overtime.

In a study by Mansur and Elyasiani (1994) documented that changes in the intensity of financial regulation on the banking industry may affect beta's of the banking stocks. Similar view was held by Dickens and Philippatos (1994) in American Banking sector and Brooks and Faff (1995) in Australian Banking industry.

Bal Krishna and Rekha tried to investigate that whether betas of stocks are close to market beta i.e. one and stationarity of beta for a period of twelve years and for different investment horizon. The study revealed that 54.50 % companies show beta close to one. However, less than fifty percent portfolios (34.52 %) betas show significant difference. Even, they concluded that the beta is not stable over the study period.

Dubey (2014) pointed out that the bull and bear market influence the pattern of beta. It indicates that analysis of beta series help to know the characteristics of the bull and bearish markets. Further, Harish & Mallikarjunappa (2015) studied the Indian market microstructure and found that changes in market microstructure impacts the trading process, returns and risk (beta series) of the securities. These points inherently motivate to analyze the structure of the beta series of the Indian market.

The current paper organizes further as follows. In section three a brief discussion about the methodology employed to compute the beta and construction of portfolios along with that of various models used to test the stability of betas have been discussed. Section four provides empirical results and the final section presents the discussion and conclusion and the results have been compared to the possible evidences.

III. METHODOLOGY

The CNX Nifty or Nifty 50 is a robust diversified 50 stock index constituting 23 sectors of the Indian economy. It is the largest individual financial product in India. It is used for wide range of activities such as index based derivatives trading, hedging, OTC derivatives, ETF (both onshore and offshore), fund portfolios. Since CNX Nifty 50 is much more stable than any other indices in India, the current study on stability of beta has been undertaken of the CNX Nifty 50.

For the purpose of analysis, the daily data from 1st of April 2000 to 31st March 2015 is collected from capital line data base. Based on market capitalization, five portfolios of ten companies each has been constructed. The portfolio 1 comprises top ten market capitalization stocks and portfolio 2 compose the next top M cap stocks, portfolio 3 involves the next top ten, fourth portfolio constitute the next and the last portfolio that is portfolio five was formed with the help of the last ten that is the lowest market capitalization stocks out of CNX Nifty 50 stocks. Later various stability hypotheses have been tested on individual stock betas and portfolios.

Daily Returns

The returns used in this study consist of daily stock and index returns using adjusted closing price of the CNX Nifty 50 index from 1st of April 2000 to 31stMarch 2015. The daily returns (R_t) computed from CNX Nifty 50 index by applying continuously compounded returns of the data series as follows.

$$R_t = ln\left[\frac{C_t}{C_{t-1}}\right] \quad \ldots\ldots\ldots\ldots\ldots(1)$$

Where: R_t *= return on day 't'*
C_t *= Closing Price on day 't'*
C_{t-1} *= Closing Price on day't-1'*
And ln = natural log of underlying market series.

BETA computation

The regression equation used to compute beta values for each of the 50 scripts for the period is:

$$R_{it} = \alpha_i + \beta_i R_{mt} + e_{it}$$

Where,
R_{mt} indicates the rate of return on the CNX NIFTY (Market return) for time period *'t'*
α_i and β_i are the regression parameters to be estimated, α_i is the intercept of the equation or constant term and β_i beta value of the security or market sensitivity of the stock (is the regression coefficient)
R_{it} is the individual security return for time period *'t'*
e_{it} is the error term

Chow Breakpoint test

One of the major assumptions about a time series distribution is that the underlying process is the same across all information in the distribution. A time series data may include a structural break; due to an abrupt change in policy of the state or a sudden shock in the economy, for example subprime crisis of 2008. A devise that is exceptionally useful in this regard is the Chow breakpoint test. Therefore, in order to investigate for the structural break in the time series, we use the Chow test. The following is the formula to compute the structural break points by using Chow test

$$F = \frac{(S_C - (S_1 + S_2))/(k)}{(S_1 + S_2)/(N_1 + N_2 - 2k)}$$

Where, S_C is the sum of squared residuals from the entire data. S_1 is the sum of squared residuals of the first interval (before 2008 crisis) and S_2 is the sum of squared residuals from the second interval i.e. after subprime crisis. However, $N_1 + N_2$ stand for the number of observations from first interval and second interval respectively. *k* is the total number of assigned parameters for the purpose of the study. The equation is an application of F-test, and it needed the sum of squared

errors from three regressions one for the pooled data and one for each sample period
The following two hypotheses have been framed:

The null hypothesis of the test is that, there are no structural break points in the data.

H_0: $a_1 = a_2$

H_1: $a_1 \neq a_2$

a_1 = beta series before 2008
a_2= beta series after 2008

Multiple breakpoint tests

Bai (1997) showed a new methodology to identify the breaks and further, the study conducted by Bai and Perron (1998, 2003a) created the methodology to identify the multiple breaks in the series. In this study, we estimate the structural breaks in the context of Global Maximize Test (GMT) created by Bai and Perron (1998). Further, we use U Dmax and W Dmax to estimate the multiple breaks in the series. These two factors help to estimate the breaks by maximizing the breakpoints and incorporating the weights to the individual company betas.

CUSUM test (Cumulative Sum test)

The CUSUM test is a sequential analysis used to investigate the sequential flow of residuals. This test works based on the unique graphical representation by plotting the series and this test also constructs the two critical lines at 5% probability. From this we can assess the stability of the series.
The test statistic is

$$\tilde{e}_t = \frac{h_t - x_t{}^T \hat{\beta}^{(i-1)}}{\sqrt{1 + x_t{}^T \left(X^{(t-1)}{}^T X^{(t-1)}\right)^{-1} x_t}}$$

Where, e represents the residual series, $t = v + 1, \dots\dots\dots\dots, n)$ and $h_t$denotes the dependent variable. $x_t$signs$(1, x_{t2,\dots\dots\dots,} x_{tv})^T$, It is the vector of V*1 exogenous variables. In the model, e_t is predicted to show zero mean.$X^{(t)}$shows the regressor matrix based on all observations up to i. The $\hat{\beta}^{(i)}$ is the coefficient of OLS regression based on the series of variables$(i + 1, \dots\dots\dots\dots, i + j)$

IV. DATA AND EMPIRICAL RESULTS

This section explores the structural, unknown breaks in the computed beta series of stocks listed in CNX Nifty Fifty. The following Table No.4.1 shows Chow Breakpoints statistics of CNX Nifty Fifty individual stocks.

INVESTIGATION OF STRUCTURAL BREAKS

Table No. 4. 1

CHOW TEST STATISTICS OF CNX NIFTY FIFTY STOCKS

Companies	F –Stats	P- Value	Log Likelihood Ratio	P-Value
ACC	0.203437	0.66	0.235354	0.6276
Ambuja Cements Ltd.	0.907193	0.3597	1.020293	0.3124
Asian Paints Ltd.	0.086209	0.7737	0.099144	0.7529
Axis Bank Ltd.	0.14085	0.7135	0.161645	0.6876
Bank of Baroda	0.026388	0.8735	0.030416	0.8615
Bharti Airtel Ltd.	0.456618	0.5132	0.528739	0.4671
BHEL	0.216919	0.6491	0.248226	0.6183
BPCL	0.041382	0.842	0.047673	0.8272
Cipla Ltd.	1.046771	0.3264	1.170876	0.2792
Dr. Reddy's Laboratories Ltd.	0.020521	0.8885	0.023921	0.8771
GAIL (India) Ltd.	0.048513	0.8291	0.055873	0.8131
Grasim Industries Ltd.	1.241292	0.287	1.378069	0.2404
HCL Technologies Ltd.	1.309352	0.2748	1.449844	0.2286
HDFC Bank Ltd.	0.214302	0.6511	0.245255	0.6204
Hero Moto Corp Ltd.	0.104313	0.7523	0.121173	0.7278
Hindalco Industries Ltd.	0.141151	0.7132	0.161988	0.6873
Hindustan Unilever Ltd.	0.356185	0.5617	0.409502	0.5222
Housing Development Finance Corporation Ltd.	0.230095	0.6394	0.263172	0.6079
ICICI Bank Ltd.	0.113991	0.741	0.130955	0.7174
IDFC	0.728349	0.4182	0.871347	0.3506
IndusInd Bank Ltd.	0.176114	0.6821	0.203974	0.6515
Infosys Ltd.	0.180934	0.6775	0.20733	0.6489
ITC	0.109038	0.7465	0.125289	0.7234

Jindal	0.099325	0.7576	0.11417	0.7354
Kotak Mahindra Bank Ltd.	0.136676	0.7176	0.15688	0.692
Larsen & Toubro Ltd.	1.262531	0.2815	1.390298	0.2384
Lupin Ltd.	0.55075	0.4712	0.622388	0.4302
Mahindra & Mahindra Ltd.	0.00098	0.9755	0.00113	0.9732
Maruti Suzuki India Ltd.	0.102501	0.7554	0.122375	0.7265
NTPC Ltd.	0.000246	0.9878	0.000301	0.9862
Oil & Natural Gas Corporation Ltd.	1.301454	0.2745	1.431177	0.2316
Punjab National Bank	0.010837	0.919	0.012801	0.9099
Reliance Industries Ltd.	0.082337	0.7787	0.094705	0.7583
Sesa	0.119478	0.7351	0.137229	0.7111
State Bank of India	0.031019	0.8629	0.035748	0.85
Sun Pharmaceutical Industries Ltd.	0.203199	0.6596	0.232647	0.6296
Tata Consultancy Services Ltd.	0.035633	0.8545	0.043466	0.8349
Tata Motors Ltd.	0.045585	0.8342	0.052506	0.8188
Tata Power Co. Ltd.	0.014637	0.9056	0.016879	0.8966
Tata Steel Ltd.	0.039409	0.8457	0.045403	0.8313
Ultra Tech Cement Ltd.	0.06057	0.8111	0.073783	0.7859
Wipro Ltd.	0.007439	0.9326	0.008581	0.9262
Zee Entertainment Enterprises Ltd.	0.020664	0.8879	0.023824	0.8773

It is evident from Table No.4.1 that the F-Statistics and Log likelihood ratio accept the null hypothesis of no breaks at specified breakpoints (2008 subprime crisis) for all the companies which are listed in CNX Nifty 50 Index. This shows that 2008 subprime crisis hardly had any impact on beta values of the CNX Nifty fifty stocks.

Table No. 4. 2

CHOW TEST STATISTICS OF FIVE PORTFOLIOS CONSTRUCTED BY USING CNX NIFTY FIFTY STOCKS

	Specified Break is 2008			
	F –Stats	P- Value	Log Likelihood Ratio	P-Value
Portfolio I	0.122522	0.7319	0.140709	0.7076
Portfolio II	0.004583	0.9471	0.005345	0.9417
Portfolio III	0.000861	0.9771	0.001005	0.9747
Portfolio IV	0.012745	0.9118	0.014698	0.9035
Portfolio V	0.112510	0.7431	0.130651	0.7178

In order to investigate the impact of subprime crisis 2008 on beta stability on portfolio of stocks comprising of stocks listed on CNX Nifty fifty stocks Chow test has been conducted. It is clear from table No. 4.2 that the F-Statistics and Log likelihood ratio accept the null hypothesis of no breaks at specified breakpoints (2008 subprime crisis). This indicates that 2008 subprime crisis has hardly any impact on the portfolios. Another notable observation is that the standard deviation of the entire constructed five portfolio betas are less compared to that of the individual stock betas. There is less difference in the standard deviation of portfolios in pre and post 2008 subprime crisis period compared to those comparable to individual stocks. The test seems to indicate that the portfolios are more stable than the individual stock's beta. Based on these findings, we may conclude that the construction of portfolios based on Market capitalization has lessened the 2008 crisis impact on portfolios beta series.

INVESTIGATION OF UNKNOWN BREAKS

CNX Nifty 50 stocks were not affected by 2008 subprime crisis. Further to investigate the presence of any unknown breaks in the computed beta series other than subprime crisis 2008 that may affect the stability of the betas for both individual shares and portfolios, multiple breakpoint test has been conducted.

Table No. 4. 3

MULTIPLE BREAKPOINT TEST (BAI AND PERRON) STATISTICS OF NIFTY FIFTY STOCKS

Companies	UD max determined breaks	WD max determined breaks
ACC	5	5
Ambuja Cements Ltd.	1	1
Asian Paints Ltd.	1	5
Axis Bank Ltd.	3	3
Bank of Baroda	2	5
Bharti Airtel Ltd.	5	5
BHEL	4	4
BPCL	5	5
Cipla Ltd.	5	5
Dr. Reddy's Laboratories Ltd.	4	4
GAIL (India) Ltd.	5	5
Grasim Industries Ltd.	1	5
HCL Technologies Ltd.	2	2

HDFC Bank Ltd.	1	1
Hero MotoCorp Ltd.	5	5
Hindalco Industries Ltd.	5	5
Hindustan Unilever Ltd.	1	2
Housing Development Finance Corporation Ltd.	5	5
ICICI Bank Ltd.	4	4
IndusInd Bank Ltd.	0	0
Infosys Ltd.	3	5
ITC	0	0
Jindal	5	5
Kotak Mahindra Bank Ltd.	2	5
Larsen & Toubro Ltd.	4	4
Lupin Ltd.	1	5
Mahindra & Mahindra Ltd.	3	3
NTPC Ltd.	3	3
Oil & Natural Gas Corporation Ltd.	5	5
Punjab National Bank	3	5
Reliance Industries Ltd.	2	5
Sesa	5	5
State Bank of India	3	3
Sun Pharmaceutical Industries Ltd.	0	0
Tata Motors Ltd.	2	2
Tata Power Co. Ltd.	1	5
TATA Steel Ltd.	1	4
Wipro Ltd.	4	4
Zee Entertainment Enterprises Ltd.	1	1

Table No 4.3 Presents the both with breaks and without break betas series of CNX Nifty fifty shares for the study period between 2000 to 2015 based on the UD max determined breaks and WD max determined breaks result. According to the UD max determined breaks results, 3 (7.69%) companies betas have no structural breaks, followed by, 9 (23.08%) companies betas have single breaks, 5 companies (12.82%) have double breaks in their beta series. Another 6 companies have (15.38%) triple breaks in their betas, five companies (12.82%) have four breaks in their computed betas and balance 11 companies (28.21%) have five breaks in their beta series for a study period from 2000 to 2015.

However, WD max results, 3 (7.69%) companies betas have no structural breaks, followed by, 3 (7.69%) companies' betas have single breaks, 3 companies (7.69%) have double breaks in their beta series, another four companies have (10.26%) triple breaks in their betas, six more companies (15.38%) have four breaks in their computed betas and balance 20 companies (51.28%) have five breaks in their beta series for a study period from 2000 to 2015.

The current study points out that the presence of the unknown breaks in major chunk of CNX Nifty Fifty individual stocks beta series for a study period of 2000 to 2015, though all the stocks which are listed in CNX Nifty Fifty have no structural breaks during the 2008 subprime crisis. That means 7.69 % of companies do not have any breaks in their beta series, however, balance 92.31% of the companies have breaks in their betas.

Table No. 4. 4

MULTIPLE BREAKPOINT TEST (Bai and Perron) STATISTICS OF FIVE PORTFOLIOS CONSTRUCTED BY USING NIFTY FIFTY STOCKS

Portfolio	UD max determined breaks	WD max determined breaks
Portfolio I	4	4
Portfolio II	4	5
Portfolio III	2	5
Portfolio IV	4	4
Portfolio V	5	5

The Table No. 4.4 shows the identified breaks for the constructed portfolios. The results of Bai and Perron test, portfolio 1 has four UD max determined breaks and four WD max determined breaks. Whereas portfolio two has four UD max determined breaks and five WD max determined breaks. Portfolio 3 with two UD max determined breaks and four WD max determined breaks; portfolio 4 has four UD max determined breaks and four WD max determined breaks and the last portfolio that is the portfolio 5 consists of least market capitalization category of stocks has five breaks each in the study period of 2000 to 2015.

Investigation of Sequential Breaks

In the last step CUSUM test has been conducted to find the sequential changes in the beta series of the stocks and also in the constructed portfolios. Out of stocks listed on CNX Nifty Fifty 32 (64%) companies have stable betas. However, 18 (36%) companies' betas are unstable at 5% level of significance.

Table No. 4.5

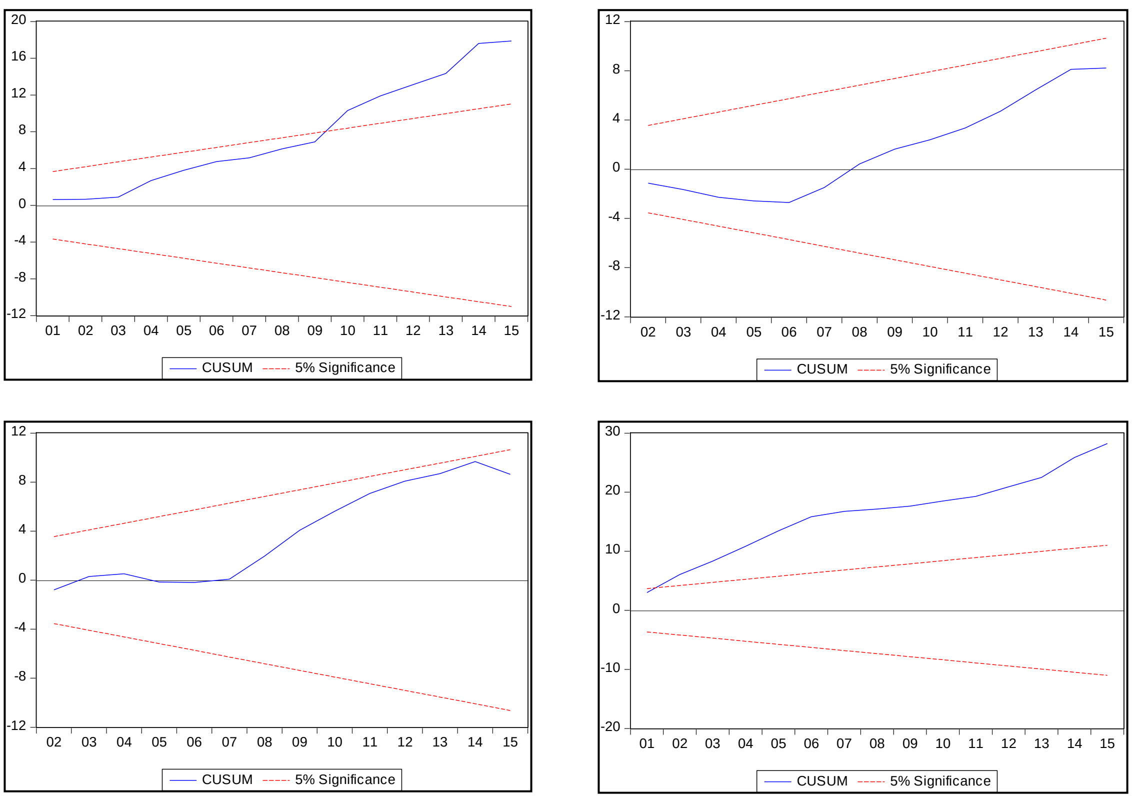
CUSUM Test Results of CNX Nifty Fifty Stocks

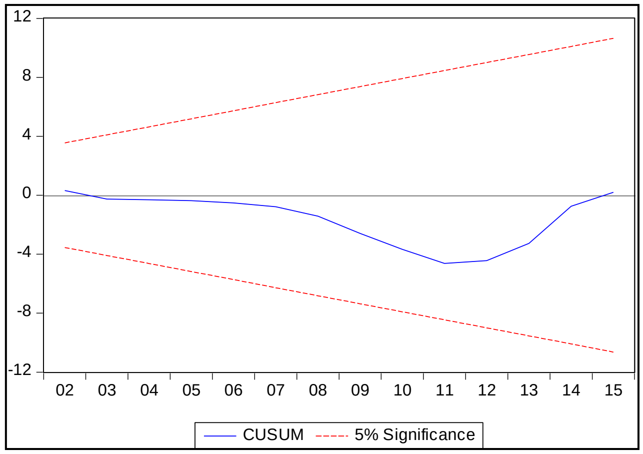
Total Number of Companies listed in the index	50
Beta stable companies	32*
Beta unstable companies	18*

*at 5% level of significance

Observe following Exhibit 4.1. As for as portfolios are concerned, the CUSUM test statistic of Portfolios-1 and Portfolios-4 are unstable and is higher than the portfolios 2, 3 and 4. However, the results for portfolio 2, 3 and 4 CUSUM test series nearer to zero mean value. The CUSUM values for portfolio 1 and 2 have more deviation compared to the portfolios 2, 3 and 4. It is interesting to note that the stocks which have more breaks as per the results of multiple break points test are the major components of portfolios 1 and 4. This may answer the major logic behind the more deviation in the portfolios 1 and 4.

EXHIBIT No. 4.1: CUSUM TEST RESULTS OF CNX NIFTY FIFTY STOCKS





V. DISCUSSION AND CONCLUSION

The study was conducted on daily data of NIFTY from 2000 to 2015 March 31st to test the stability of beta. The data was tested for structural breaks in beta series in response to subprime crisis of 2008. The study also tested for stability of BETA for unknown breaks in the series. Based on the Chow Breakpoint test we can conclude that 2008 subprime crisis has no impact on any Indian individual stocks as well as on the portfolios constructed by using CNX Nifty Fifty Stocks. However, multiple breakpoint tests revealed that there is an existence of unknown breaks in the computed beta series. Majority of the companies have more than two breaks in their beta series. The above breaks have adversely affected the stability of beta even on the constructed portfolios. All the five portfolios constructed by using market capitalization as a base evidenced a series of unknown breaks in the computed portfolios betas.

Although majority of the available literature recorded that portfolio betas are much more stable than individual stocks beta, the outcome of this study indicated that both individual as well as portfolio betas behave very similar with respect to its stability. Our results seem to agree with the results of Deb and Misra (2011), Balkrishnan and Rekha Gupta (2012), Sibel Celik (2013).

In the last phase, out of fifty stocks taken for the purpose of the study listed on CNX Nifty Fifty majority of the companies have their betas stable. Further we do not find any pattern in the stability of beta with respect to portfolios.

The present study revealed that the beta values are not stable over a period of time thus; the investors should take precaution while measuring the systematic risk. Apart from this it is strongly recommended to the market participants to consider the time-varying behavior of beta while making the investment decision on individual stocks and managing portfolio. The empirical findings of the study is very important for the market participants as beta has a predictive power of prospective risk of a stock and it helps the investor to estimate exactly the security's projected riskiness. Hence, the participants should be very careful while calculating risk by taking beta as a parameter. They should keep updating the beta while holding, buying and selling individual stocks and constructing and managing portfolios. Thus, they can alter their investment strategy based on the actual changes in the levels of beta rather than the computed beta based on the historical returns.

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